Document Clustering in NLP

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Over 2.5 Quintillion Bytes of Data Every Single Day

1.7MB of Data Every Second for Every Person on Earth

Outline

- Introduction
- Clustering
- Preprocess of Documents
- Document Clustering Challenges
- Traditional vs. Semantic Clustering
- Hierarchical Clustering Algorithems
- Partitional Clustering Algorithms
- K-means Algorithm
- Conclusion

Introduction

- Natural Language Processing (NLP)
- Application of Clustering in NLP
- Effective Clustering
- Clustering Methods

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Clustering

- Clustering Definition
- Hard Clustering vs. Soft Clustering

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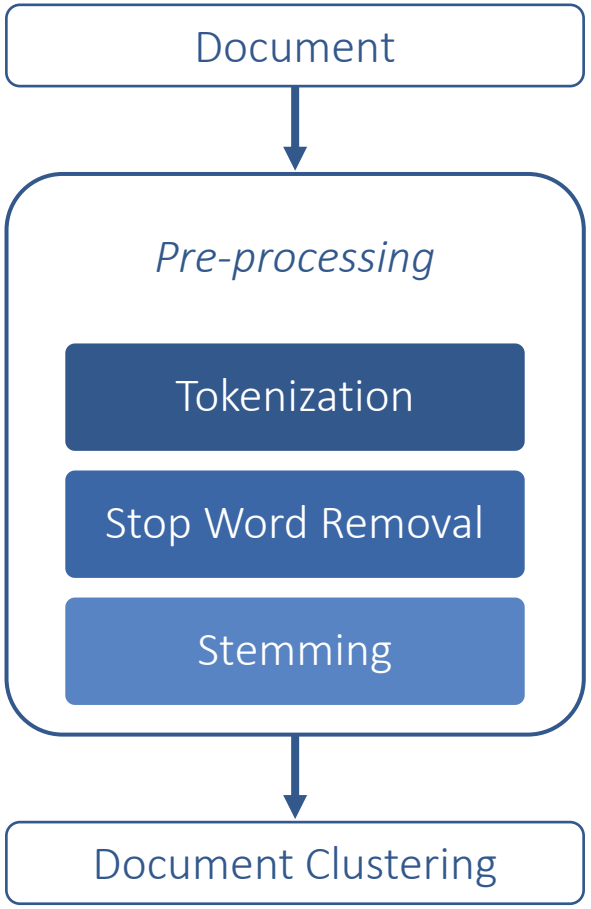
Preprocess of Documents



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Document Clustering Challenges

- Synonymy and Polysemy
- Ambiguity
- High Dimensionality
- Cluster Labeling

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Traditional vs. Semantic Clustering

- Traditional Document Clustering
- Semantic Document Clustering
- Advantages of Semantic Clustering over Traditional Clustering

Outline

- Massive Amount Of Data
- Organizing Data for Further Usages
- Clustering as an Organizing Method
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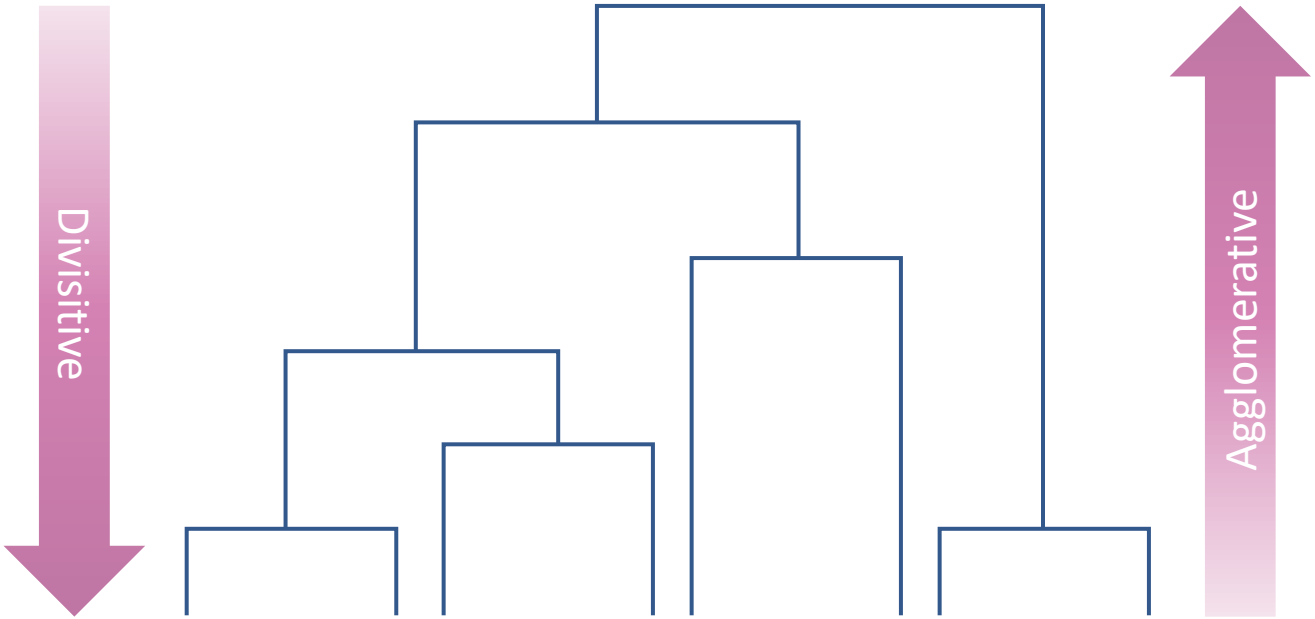
Hierarchical Clustering Algorithems

- Agglomerative Algorithms
- Divisitive Algorithms
- Advantages and Disadvantages of Hierarchical Clustering Algorithms

Agglomerative vs. Divisitive



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Patritional Clustering Algorithems

- Direct Partitional Algorithms
- Bisecting Partitional Algorithms

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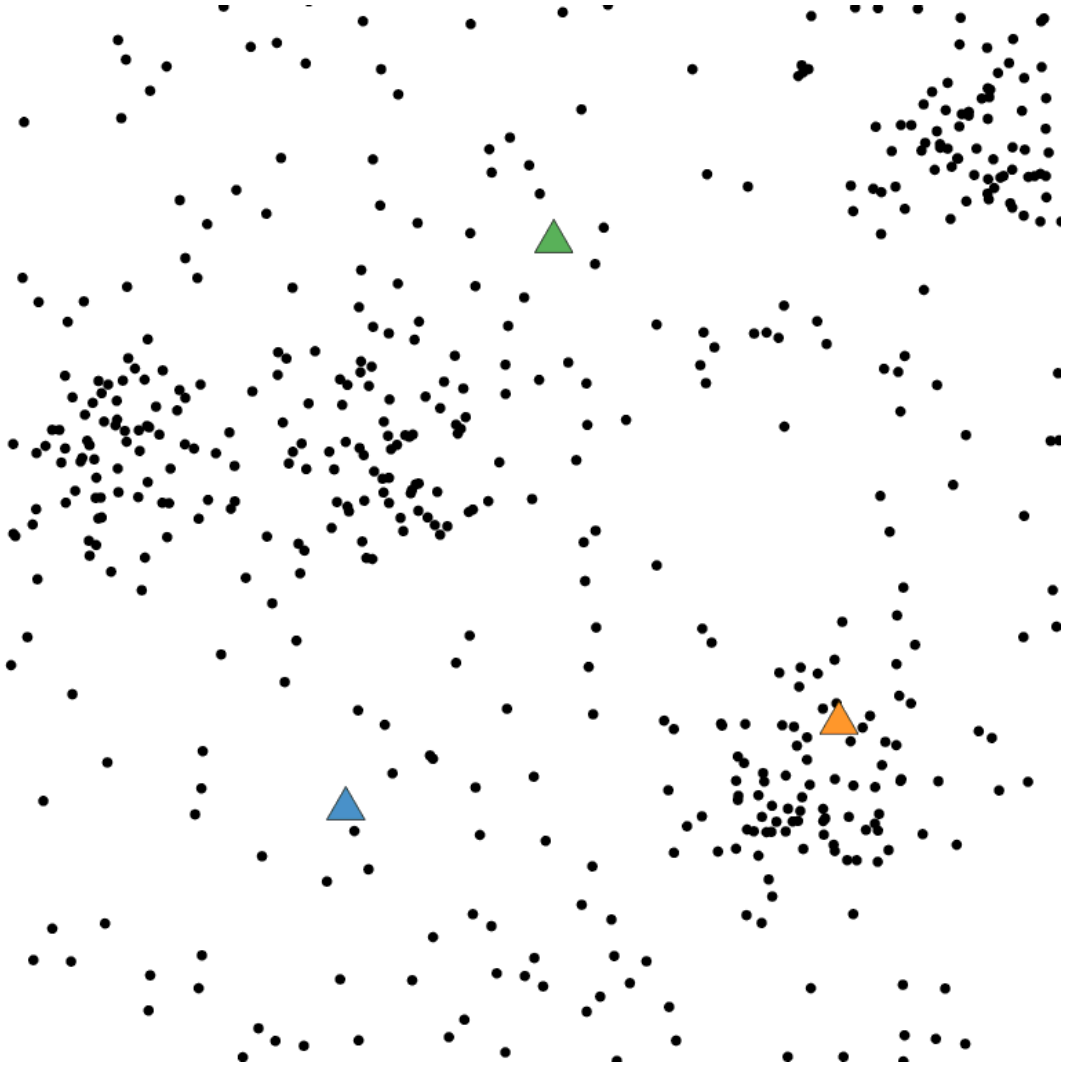
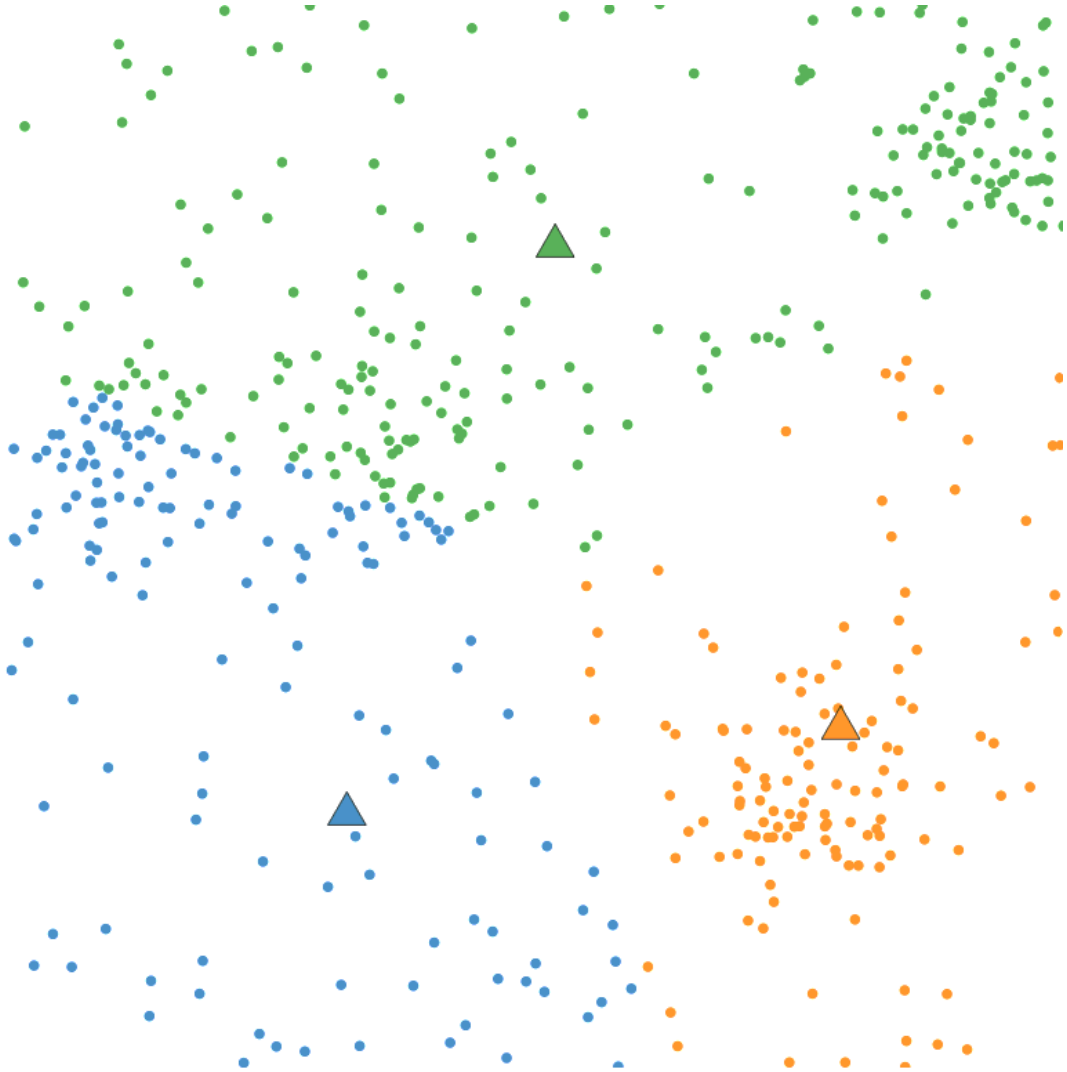
K-means Algorithem

- How It Works
- Choosing Centroids
- Objective Function

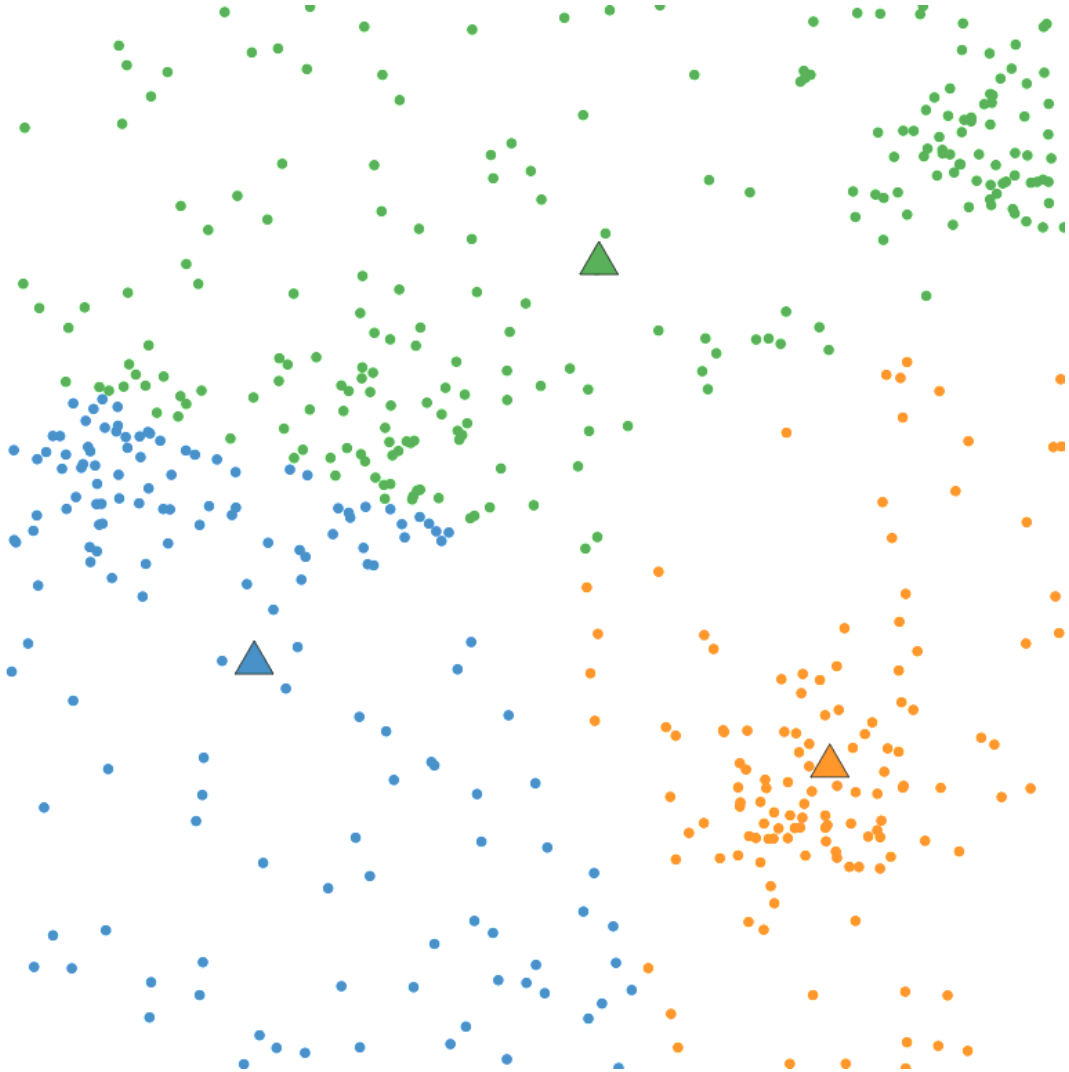
K-means: Initiate Centroids



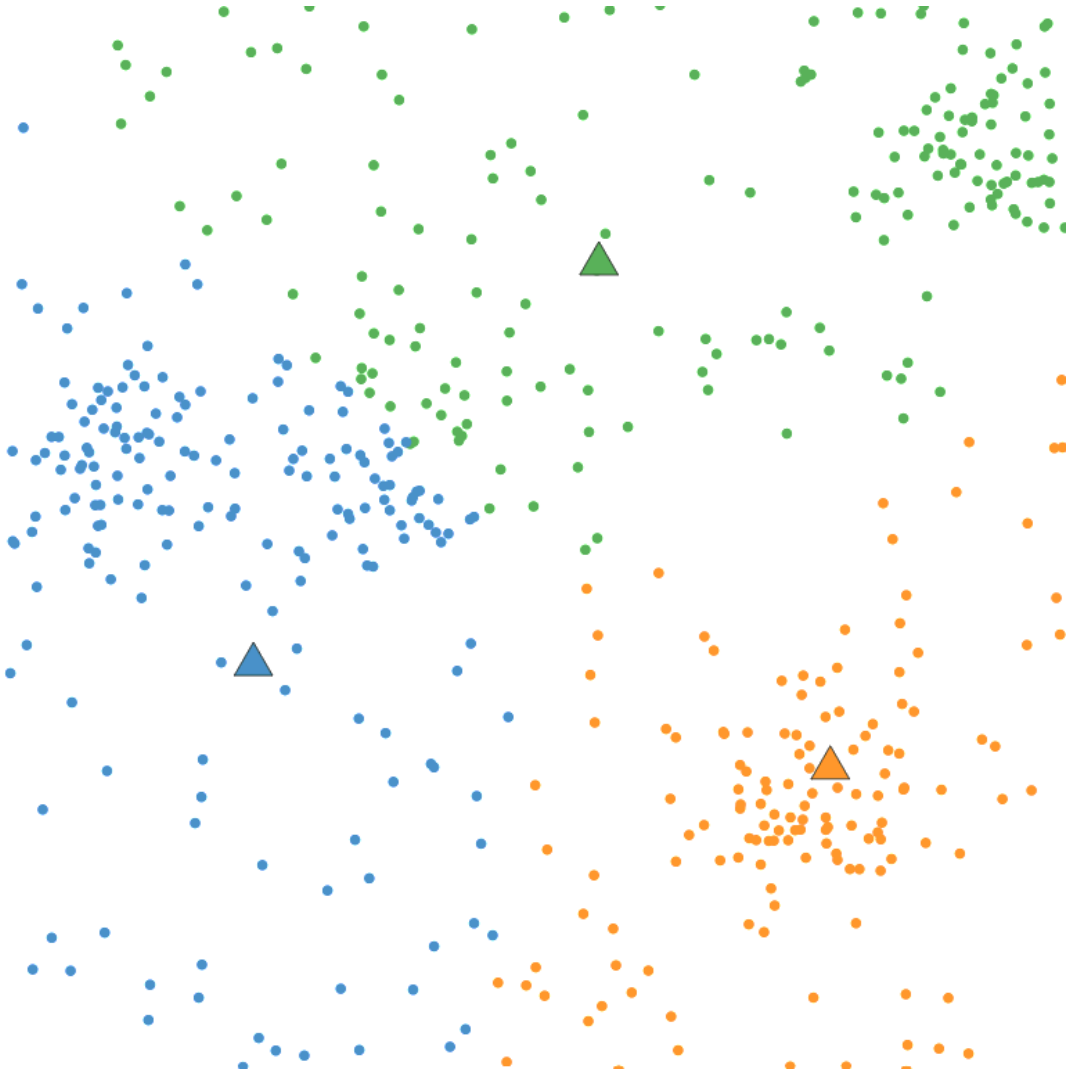
K-means: Find Closest Centroids



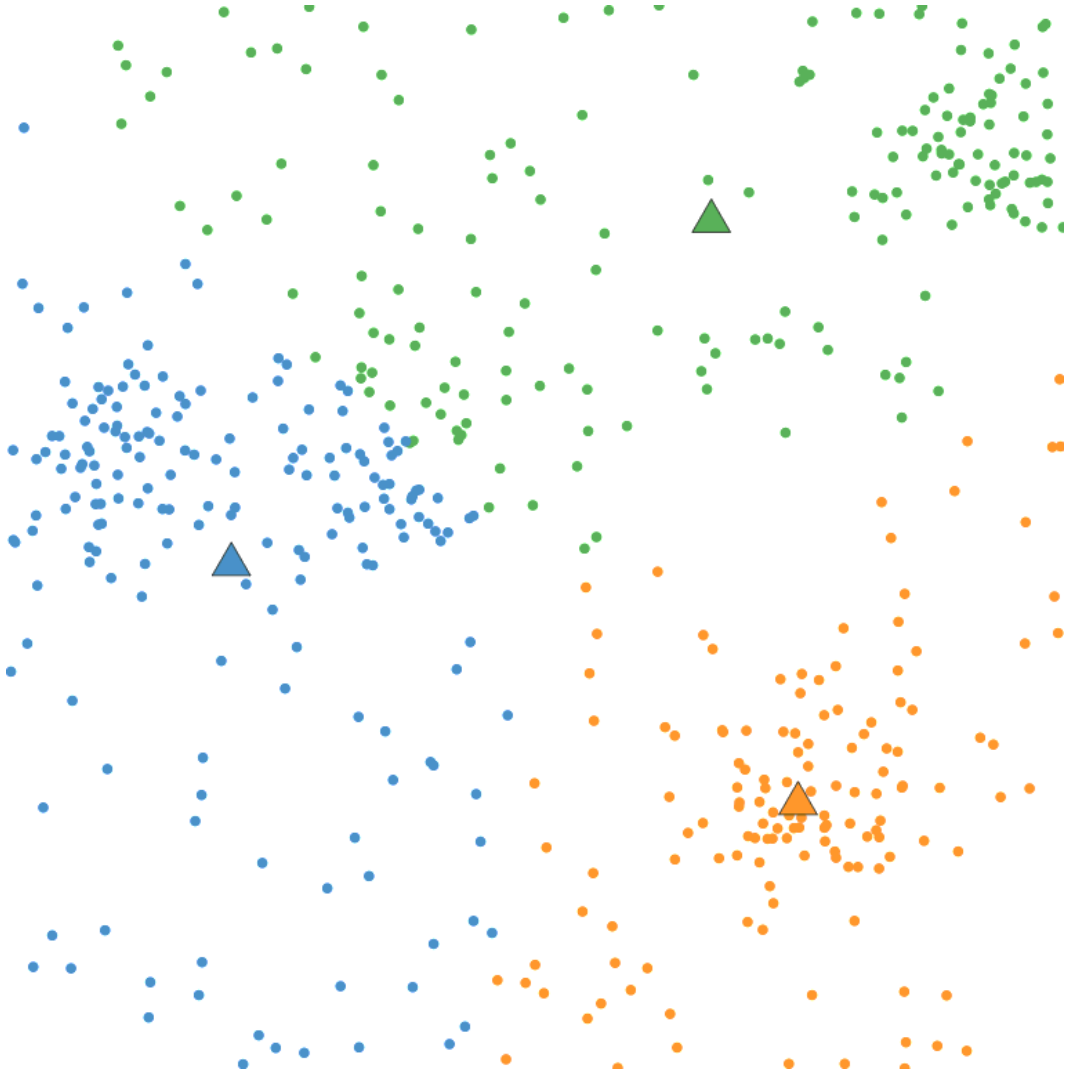
K-means: Update Centroids



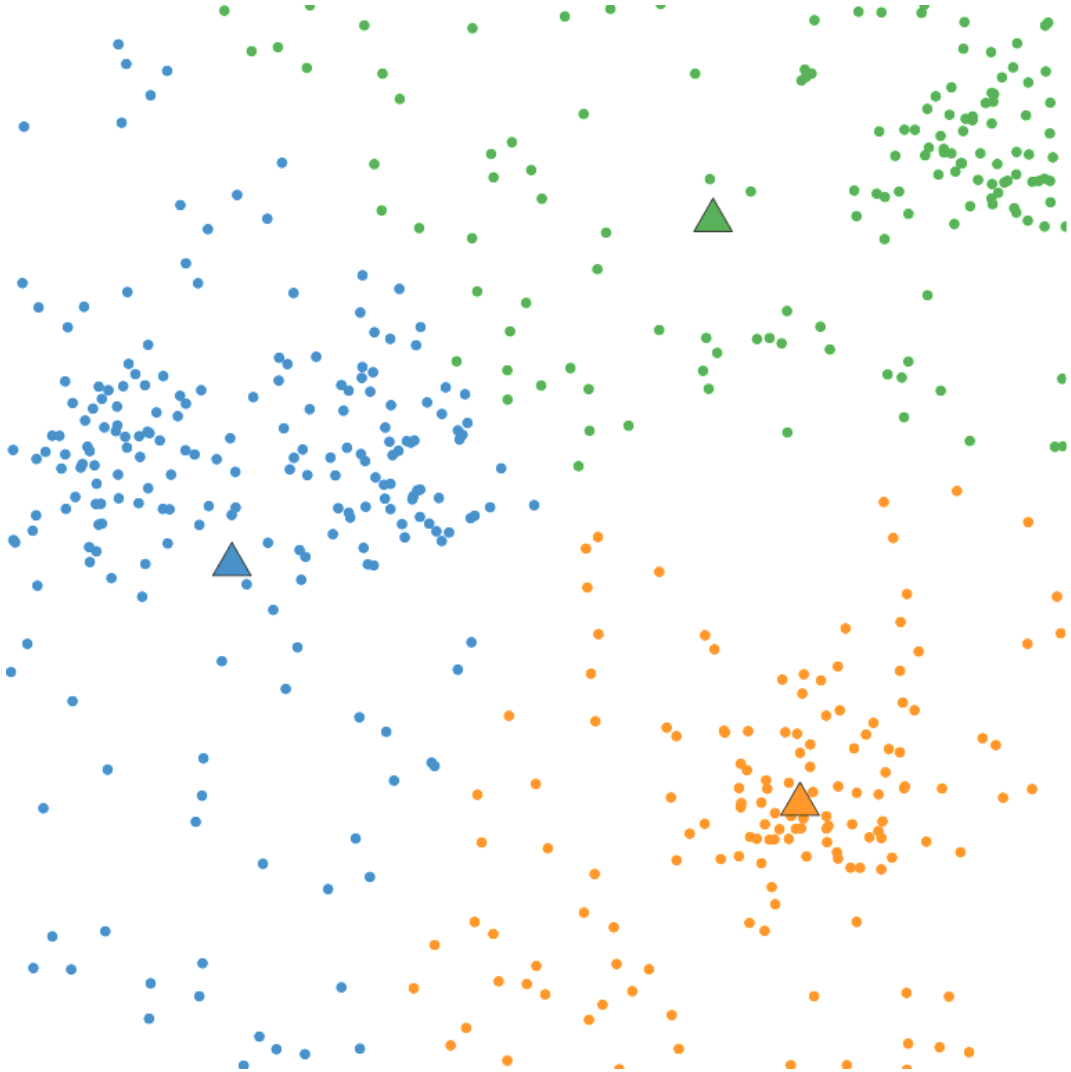
K-means: Find Closest Centroids



K-means: Update Centroids



K-means: Find Closest Centroids



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Conclusion

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- Meaningfully Organizing Data for Further Usages
- Clustering as an Organizing Method